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27 June 1949

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

STEEL, SPRING, ROUND OR SQUARE (INCLUDING RECTANGULAR SECTIONS HAVING THICKNESSES GREATER THAN HALF THEIR WIDTH)

This specification was approved by the Departments of the Army, Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Navy 47S4g
1 September 1943

This specification consists of this cover sheet and Specification 47S4g dated 1 September 1943, attached hereto, without modification.

Copies of this specification may be obtained from the Bureau of Supplies and Accounts, Navy Department, Washington 25, D.C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

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47S4g
SEPT. 1, 1943
SUPERSEDING
47S4f
May 1, 1941

NAVY DEPARTMENT SPECIFICATION

STEEL, SPRING, ROUND OR SQUARE (INCLUDING RECTANGULAR SECTIONS HAVING THICKNESSES GREATER THAN HALF THEIR WIDTH)

A. APPLICABLE SPECIFICATIONS.

A–1. The following Navy Department specifications, of the issue in effect on date of invitation for bids, form a part of this specification, and bidders and contractors should provide themselves with the necessary copies:

General Specifications for Inspection of Material, together with Appendix II, Metals, Part A, Definitions and Tests.
22W11—Wire, steel (high-carbon), spring (music wire).

B. GRADES, TYPES, AND COMPOSITIONS.

B–1. Spring steel covered by this specification shall be of the respective grades, types, and compositions specified in tables I and II. Grades and types are defined as follows:

B–1a. Grade I spring steel consists of products which, within the respective ranges of sizes prescribed for each in table I, are considered to be the best materials commonly available for the manufacture of springs to meet exacting requirements.

B–1b. Grade II material is a lower grade than grade I material because of less exacting requirements as to finish, state, and size.

B–1c. Type A material consists of stock prepared for fashioning while cold into springs which require no subsequent hardening.

B–1d. Type B material consists of stock for the manufacture of springs which are to be hardened (and tempered) after hot or cold forming.

C. MATERIAL AND WORKMANSHIP.

C–1. *Material.*—

C–1a. Spring steel shall be made by the electric, acid open-hearth, or basic open-hearth process, unless a particular process is specified.

C–1b. Spring steel designated in this specification as wire shall be the product made from hot-rolled rods by cold drawing through dies.

C–1c. Spring steel designated in this specification as bars shall be produced by hot rolling, with or without a final cold rolling or cold drawing.

C–1d. Except as permitted by the provisions of sections E and F of this specification, spring steel shall be free from all internal or surface defects or decarburization which might affect its serviceability.

C–2. *Workmanship.*—The workmanship shall be first class.

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D. GENERAL REQUIREMENTS.

D–1. There are no general requirements applicable to this specification.

E. DETAIL REQUIREMENTS.

E–1. *Size of section.*—The size of section is a major consideration in the use of spring steels. The materials listed in table I are acceptable only within the ranges specified herewith.

TABLE I.—*Classification of spring steel, round or square*

Material number	Type and name of material	Composition number	Size limits, diameter, or width (specified or nominal)	
			Minimum	Maximum
			Inch	*Inches*
	Grade II, type A.—Spring temper wire:		0.004	[1] 0.156
1	Music wire, Navy Department Specification 22W11	2	[2] .032	.282
2	Tempered, medium-carbon	3	.220	.500
3	Tempered, medium-carbon manganese	3	[2] .028	.500
4	Hard drawn wire			
	Grade II, type B.—Annealed wire and bars:	1		.500
5	High-carbon wire, round and square	1	.250	1.250
6	High-carbon square bars	1	.375	1.250
7	High-carbon round bars	2	[2] .032	.282
8	Medium-carbon wire, round and square	3	.220	.500
9	Medium-carbon manganese wire, round and square	3	.250	.750
10	Medium-carbon manganese square bars	3	.375	.750
11	Medium-carbon manganese round bars	5	.187	
15	Silicon manganese, centerless ground round bars	5	.375	
16	Silicon manganese hot-rolled bars, round and square			

[1] 0.156 inch is the commercial limit. Any size covered by Navy Department Specification 22W11 is acceptable.

[2] Where minimum sizes less than 0.050 inch are given in the table, smaller sizes are acceptable if obtainable.

NOTE.—Grade I shall be the same as grade II with the following additional limitations:

(*a*) Hard-drawn wire, material 4, is not acceptable.

(*b*) Music wire only, material 1, is acceptable as round wire smaller than 0.10 inch in diameter.

(*c*) Type B square wire is not acceptable in sizes less than 0.10 inch thick.

(*d*) Type A wire is not acceptable in sizes greater than 0.375 inch in diameter or width.

(*e*) Hot-rolled round bars of composition 5 are not acceptable in diameters less than 0.50 inch.

E–2. *Chemical composition.*—The chemical composition shall be as shown in table II.

TABLE II.—*Chemical composition*

Number	Name	Carbon	Manganese	Silicon	Phosphorus (maximum)	Sulphur (maximum)
		Percent	*Percent*	*Percent*	*Percent*	*Percent*
1	High-carbon	0.90–1.05	0.25–0.50	0.10–0.50	0.045	0.050
2	Medium-carbon	.55–.75	.60–.90	.10–.30	.045	.050
3	Medium-carbon manganese	[1] .55–.75	.80–1.20	.10–.30	.045	.050
5	Silicon manganese	.50–.65	.60–.90	1.80–2.20	.045	.050

[1] For hard drawn wire smaller than 0.098 inch in diameter or width, a minimum carbon of 0.45 percent is acceptable.

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E–3. *Physical properties.*—

E–3a. Music wire shall conform to Navy Department Specification 22W11, listed in section A.

E–3b. Tempered spring wire, materials 2 and 3, shall have the tensile strength shown in table III.

TABLE III.—*Tensile strength, tempered spring wire, round or square*

Nominal diameter or thickness (inch)	Tensile strength (pounds per square inch)		Nominal diameter or thickness (inch)	Tensile strength (pounds per square inch)	
	Minimum	Maximum		Minimum	Maximum
0.032	275,000	310,000	0.148	205,000	235,000
0.035	268,000	303,000	0.162	200,000	230,000
0.041	261,000	296,000	0.177	195,000	225,000
0.047	254,000	289,000	0.192	190,000	220,000
0.054	248,000	283,000	0.207	185,000	215,000
0.063	242,000	277,000	0.225	183,000	213,000
0.072	236,000	271,000	0.244	182,000	212,000
0.080	230,000	265,000	0.250	180,000	210,000
0.093	225,000	260,000	0.312	172,000	208,000
0.105	220,000	255,000	0.375	175,000	205,000
0.120	215,000	250,000	0.437	170,000	200,000
0.135	210,000	240,000	0.500	165,000	195,000

NOTE.—The requirements for intermediate sizes shall be obtained by interpolation.

E–3c. Hard-drawn spring wire, material 4, shall have the tensile strength specified in table IV.

TABLE IV.—*Tensile strength, hard-drawn spring wire*

Nominal diameter or thickness (inch)	Tensile strength (pounds per square inch)		Nominal diameter or thickness (inch)	Tensile strength (pounds per square inch)	
	Minimum	Maximum		Minimum	Maximum
0.028	260,000	310,000	0.135	190,000	231,000
0.032	253,000	300,000	0.148	186,000	228,000
0.035	250,000	297,000	0.162	182,000	223,000
0.041	240,000	288,000	0.177	180,000	220,000
0.047	235,000	280,000	0.192	176,000	217,000
0.054	230,000	274,000	0.207	172,000	214,000
0.063	220,000	268,000	0.225	170,000	211,000
0.072	215,000	260,000	0.244	168,000	209,000
0.080	210,000	254,000	0.250	167,000	208,000
0.093	203,000	249,000	0.312	160,000	200,000
0.105	200,000	242,000	Over 0.312	150,000	190,000
0.120	195,000	237,000			

NOTE.—The requirements for intermediate sizes shall be obtained by interpolation.

E–3d. *Coiling properties.*—All wire less than 0.17 inch in diameter shall be capable of being coiled on itself as an arbor; wire 0.17 to ½ inch, inclusive, shall be capable of withstanding coiling about an arbor twice the diameter of the wire.

E–4. *Properties of type B spring stock.*—

E–4a. *Annealing.*—Type B spring steel shall be furnished in a comparatively soft condition. The condition produced by a properly restrained cooling of hot-rolled steel from the rolling temperature is acceptable provided the resulting condition is suitable for such shearing, machining, and bending operations as are ordinarily performed on the section in question. Except where fully annealed material is specified, type B spring steel shall be "annealed" by heating to a temperature

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and for a time insufficient to cause complete recrystallization. The structure may be fine-grained pearlitic, sorbitic, or partially or completely spheroidized. Annealed spring wire shall receive at least one pass—preferably two passes—through dies after the last heating. The word "annealed" as applied to spring steel under this specification shall have the significance indicated above. The definition of "annealing" in Appendix II, Metals, listed in section A, does not apply.

E–4b. Annealed medium carbon and medium carbon manganese wire, materials 8 and 9, shall have a minimum tensile strength of 100,000 pounds per square inch. No quenching cracks shall result from quenching this wire in oil from 1,575° F.; and, after sandblasting to remove scale, the wire shall show the hardness results shown in table V when tested without removing any metal to provide a flat spot.

TABLE V.—*Rockwell hardness of materials 8 and 9 after oil quench*

Diameter or thickness (inch)	Hardness, Rockwell C
	Minimum
Less than 5/16	53
5/16 to 1 5/32, inclusive	55
Over 1 5/32 to 5/8, inclusive	57

E–4c. Bar stock of composition 5, materials 15 and 16, shall be furnished in the fully annealed condition. Specimens tested after being quenched in oil and suitably tempered shall show a tensile strength of not less than 200,000 pounds per square inch with an elongation of not less than 6 percent in 2 inches. The temperature from which the specimens are quenched shall be between 1,725° and 1,750° F.

E–5. *Uniformity of temper and form.*—Each piece or coil of spring steel stock shall be of uniform temper throughout. It shall be regular in form and not crooked, buckled, wavy, or twisted.

E–6. *Twist of type A wire.*—Type A spring wire shall not have been twisted in drawing and coiling.

E–7. *Surface.*—All spring wire shall be free from checks, seams, deep scratches, and other injurious defects as judged after etching for 15 minutes in hot hydrochloric acid of 1.10 specific gravity, as specified in section F.

E–8. *Finish.*—

E–8a. Hard-drawn spring wire and annealed wire shall be furnished with a bright, smooth finish, free from scale and rust.

E–8b. Tempered spring wire shall be furnished, as specified, in the following finishes:

(1) Bright, polished.
(2) Blued.
(3) Black, the finish obtained directly from the hardening and tempering operations.

E–8c. The centerless grinding of material 15, composition 5, shall effect the removal of all decarburized surface and surface defects. The final finish shall be smooth and free from scratches.

E–8d. Hot-rolled bars shall be free from checks, scores, objectionable tool marks, and excessive scale.

E–8e. Spring steel bars, materials 6, 7, 10, 11, and 16, shall not be decarburized, partially or completely, to depths greater than 0.006 inch, plus 0.004 inch for each ½ inch or fraction thereof of (nominal) diameter, thickness, or width on which the decarburization occurs.

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E–9. *Size variation limits.*

E–9a. The tolerance for diameter or thickness of spring steel wire, whether annealed, tempered, or hard-drawn, shall be as shown in table VI.

TABLE VI.—*Tolerances for wire*

Diameter or thickness (*inch*)	Tolerance (plus or minus)
	Inch
Less than 0.072	0.001
0.073 to 0.375, inclusive	.002
0.376 and over	.003

NOTE.—For round stock, the tolerance for out of round shall be one-half the tolerance for diameter.

E–9b. The tolerances for the diameter, thickness, and width of hot-rolled rounds and rectangular sections having thicknesses greater than one-half their width are as shown in table VII.

TABLE VII.—*Tolerance for hot-rolled bars*

Dimension (inches)	Tolerance		
	Under	Over	Out of round or square
	Inch	*Inch*	*Inch*
5/16 to 5/8 inclusive	0.008	0.008	0.012
Over 5/8 to 7/8, inclusive	.009	.009	.014
Over 7/8 to 1, inclusive	.010	.010	.015
Over 1 to 1 1/4, inclusive	.012	.012	.016
Over 1 1/4 to 1 1/4, inclusive	.014	.014	.018
Over 1 1/4 to 1 1/2, inclusive	.018	.018	.022
Over 1 1/2 to 2, inclusive	.022	.022	.030

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1 *Lots.*—A lot shall consist of all material of the same section and size from one melt (if known) submitted for inspection at one time.

F–2. Spring steel shall be carefully inspected to determine conformity with this specification with respect to dimensions, finish, surface defects, etc.

F–3. *Sampling.*—If the spring steel is furnished in coils, one sample shall be taken from each coil of tempered spring wire and not fewer than one from each five coils or fraction thereof of hard-drawn wire and type B spring steel. If the material is furnished in straight lengths, not less than one sample shall be taken from each ten lengths or fraction thereof. If the material is furnished as springs, one or more samples shall be taken from each fifty or fraction thereof. At least one sample shall be taken from each item consisting of material of the same section, type, and grade in a shipment.

F–4. *Chemical analysis.*—One or more samples for chemical analysis shall be selected from each lot. If the material cannot be identified as to melt, the sample for chemical analysis shall be a composite sample taken from the fragments or left-over metal from the samples taken to secure physical test specimens.

F–5. *Physical tests and examinations.*—

F–5a. *Tension tests.*—Tension tests shall be made on samples of

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spring steel of type A, and of annealed wire, materials 8 and 9. Tension tests shall be made on samples of stock of composition 5, after hardening. Wire shall be tested in full section; bars larger than ½-inch diameter may be machined. All tests and machining shall be as specified in Appendix II, Metals, Part A, Definitions and Tests, listed in section A.

F–5b. Annealed medium carbon and medium carbon manganese wire stock, materials 8 and 9, shall be examined for cracks and tested for hardness after quenching as specified in paragraph E–4b.

F–5c. *Coiling test.*—Round wire stock, small enough to be wound by hand, may be tested by bending the end of the wire over itself to form a loop with ends crossing at right angles, the shorter end being several inches long. Then wind one end five complete turns tightly about the other with turns close together. For stiffer wire, or optionally on any size, the winding may be done by any hand or power device which will coil the wire closely about a mandrel of the required diameter for five complete turns without defacing the wire. Any break or checking of the wire after the first complete turn about itself or about the mandrel has been formed will constitute failure of the sample. Since the wire may be bent locally to a radius less than that of the wire or mandrel in forming the first turn, the attempt shall be repeated if the wire breaks before the first turn is completed. A failure in the first turn shall not count against the sample if a succeeding attempt is successful. Square wire shall be wound about a round arbor. For wire less than ¼-inch square the diameter shall be the same as, and for sizes ¼-inch square and larger the diameter shall be twice, the diagonal of the square.

F–5d. *Magnetic test.*—All spring steel of circular cross section for use by the Bureau of Aeronautics, and for other bureaus when specified in the contract or order, shall be tested by the following method. The material may be straight or coiled. If coiled, the coils shall be spaced apart at least ten diameters of the wire except in tests of springs made up with closer coiling. The surface of the material shall be clean at the beginning of the test. The wire or bar shall be magnetized circularly by direct electric current passed longitudinally through the portion under examination. The amperage employed shall be sufficient to cause the sample to become warm but not so great as to heat the metal to the boiling point of water during the period of the test. While the current is flowing, the sample, if bright, shall be immersed in, sprayed with, or flooded with kerosene or a very thin oil (or a mixture of the two) containing 1 ounce per gallon of black magnetic powder. If the sample be black or blued, it shall be dusted with a dry fine gray magnetic powder and subsequently jarred or subjected to a gentle current of air, to remove excess powder while the current is flowing. The magnetic powder will indicate any longitudinal discontinuities of the surface such as seams, cracks, or scores. Where a serious defect is indicated it shall be examined by the method described in paragraph F–5e. The magnetic test is nondestructive and permits the expeditious examination of a large sample. Its purpose is to detect and locate defects or to establish their absence. Acceptance or rejection will depend upon determination of the nature and seriousness of the defects discovered, as indicated in paragraph F–5e.

F–5e. *Hot acid etch.*—All lots of spring steel on which the absence of surface defects has not been demonstrated by the magnetic test shall be subjected to the hot acid etch test. This shall consist of immersing

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a piece, approximately 3 inches long, cut from each sample taken in accordance with paragraph F–3, for 15 minutes at about 180° to 210° F. in a 1 to 1 mixture by volume of hydrochloric acid and water. After etching, hot-rolled steel shall show no checks, seams, or sharp-bottomed scratches. Wire shall show no hair line or other cracks, seams, or other than superficial round-bottomed grooves, when examined after etching. The ends of all samples, except fine wires, shall be ground smooth before etching so as to permit the etching to reveal internal defects.

F–5f. *Microscopic examination.*—When so specified in the contract or order, and when centerless ground stock is specified, samples shall be examined microscopically for decarburization and other defects by a competent metallographist. Cross sections shall be polished and etched so as to permit examination to the extreme edge. Wire shall not be considered acceptable if decarburized (partially or completely) to a depth exceeding 0.7 percent of the diameter or thickness. On centerless ground or machined stock there shall be no surface decarburization.

F–5g. *Internal defects.*—Pipes, internal checks, or bursts shall be cause for rejection, whether found on microscopic examination, by the hot acid etch, or otherwise.

F–5h. *Test for twist and distortion.*—In taking samples of type A wire as called for by paragraph F–3, six turns shall be cut from each bundle and dropped on a flat surface. The wire shall lie flat and not spring up or show a wavy condition.

F–6. When spring steel under this specification is furnished as completed springs, inspection of the material under this specification, as distinguished from inspection of the springs as such, shall consist of examination for conformity to section C and to paragraphs E–1, E–2, E–7, and E–8. The requirements of paragraphs E–4, E–5, E–6, and E–9 concern characteristics of spring stock which relate to the operations of spring making rather than to the properties of the material after completion of the springs. Tests to determine conformity to the requirements of paragraph E–3 shall be omitted when tests of the spring are required.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Packaging (for domestic shipment only).*—Commercial packaging is acceptable under this specification. All material shall be properly separated by size.

G–2. *Packing (for domestic shipment only).*—Unless otherwise specified, the subject commodity shall be delivered in substantial commercial containers of the size and type commonly used, so constructed as to insure safe delivery by common or other carrier to the point of delivery at the lowest rate. All containers shall comply with the requirements of Consolidated Freight Classification Rules in effect at the time of shipment. Whenever practicable, a single container, when packed for shipment, shall weigh not in excess of approximately 300 pounds gross. Any special crating or boxing necessary for acceptance by commercial carriers shall be performed by the contractor without expense to the Government. Not more than one size or kind of material shall be packed in a single container.

G–3. *Marking.*—

G–3a. *Packages.*—Packages shall be marked with the name of the material, the grade, the type, the material number, the composition,

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and the quantity contained therein, as defined by the contract or order under which shipment is made, and the name of the manufacturer.

G–3b. *Shipping containers.*—Shipping containers shall be marked with the name of the material, the grade, the type, the material number, the composition, and the quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, the number of the contract or order, and the gross weight.

H. NOTES.

H–1. Requests, requisitions, schedules, and contracts or orders must contain the title of the specification, the number, and date.

H–2. This specification covers spring steel for all applications in which high temperatures, corrosion, and other special considerations are not major factors. The size limitations shown in table I do not indicate limiting sizes of material procurable, but designate ranges within which the properties of the material comply with the standards for the stated types and grades.

H–3. Requests, requisitions, schedules, and contracts or orders for material for the manufacture of springs should specify the grade, the type, and the material number; if tempered spring wire is ordered, a bright blued, or black finish should be specified. Microscopic inspection should be specified if such examination is desired.

H–4. For any given section of stock, springs, made of any of the optional types, products, and compositions under grade I ordinarily will be produced with substantially the same spring properties. Ordinarily the best springs as to uniformity and dependabilty are produced from type A material in the smaller sizes where its use is practicable. Where the required forming operations cannot be performed on type A, where apparatus adequate to form it is not available, where greater hardness than that of type A is required to meet load requirements (at the sacrifice of toughness), and in other instances, it becomes necessary or preferable to use type B. Since such considerations are not always within the cognizance of the designer and since ordinarily no advantage is gained by specifying one of these products or compositions to the exclusion of others, it is usually advisable to specify the grade only, when this specification is used to designate the material of springs on drawings and in specifications.

H–5. Diameters and thicknesses of spring steel should be designated in inches and decimal fractions thereof; but in the design of springs, and selection of sizes for use, the sizes on the American Steel and Wire Company's music wire gage should be used for music wire and the Washburn and Moen gage for tempered, annealed, and hard-drawn spring wire, if practicable. Half sizes intermediate between gage sizes on the Washburn and Moen gage may be obtained commercially.

H–6. Bars for uses in which decarburization to the depth indicated in paragraph E–8e is not permissible should be centerless ground.

H–7. Spring strip and flat stock formerly covered by Navy Department Specification 47S4 are now covered by Navy Department Specification 47S27, Steel, Spring, Flat Stock.

H–8. Copies of Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington, D. C., except that Naval activities should make application to the Commandant, Navy Yard, New York, N. Y. When requesting, refer to specification by both title and number.

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